

Project Proposal: Portable 3D Surface Reconstruction using Screen-Based Active Structured Light

Team Members: Promita Panja, Mihir Niravkumar Sukharamwala, Ashee Mishra
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Abstract

This project aims to engineer a portable, low-cost 3D scanning system that repurposes a standard laptop display as a programmable structured light projector and utilizes a commodity webcam for image capture. Traditional 3D reconstruction techniques often depend on costly laser hardware or passive stereo vision, which struggle with featureless surfaces. By adopting an *Active Vision* methodology, this project will implement a "Shape from Structured Light" pipeline enhanced by modern phase-unwrapping techniques. We will project a sequence of Phase-Shifted Sinusoidal patterns onto target objects, analyzing the surface deformation to compute dense 3D depth maps. Our primary objective is to demonstrate that high-fidelity, "correspondenceless" 3D reconstruction is achievable on consumer-grade hardware by optimizing algorithms for the low-luminance conditions typical of screen-based illumination, inspired by recent advancements in deep-learning-based profilometry.

1 Summary of Existing Work

The field of 3D reconstruction is generally divided into active and passive methodologies. While passive methods like multi-view stereo are popular, they degrade rapidly when applied to textureless objects (1). Active Structured Light (SL) solves this by projecting synthetic patterns to establish temporal correspondence.

Recent surveys by Li et al. (2025) highlight that while Digital Light Processing (DLP) projectors remain the industrial standard, there is a growing trend toward "ubiquitous scanning" using consumer devices (1). However, standard SL algorithms often fail with screen-based projection due to low contrast and high noise. To address this, recent research has pivoted toward Deep Learning (DL) integration. Zuo et al. (2024) demonstrated that deep networks can achieve single-shot 3D reconstruction with high accuracy, even in dynamic scenes (6). Similarly, Guo et al. (2025) introduced "inner shifting-phase encoding" to handle complex object surfaces without extensive multi-frame scanning (3).

On the algorithmic side, Phase Unwrapping (PU) remains the most critical bottleneck. Traditional spatial PU methods are prone to error propagation. Chen et al. (2024) recently proposed "Unsupervised Deep Unrolling Networks," which use physics-informed layers to unwrap phase maps robustly without ground truth data (7). Furthermore, Gontarz et al. (2023) applied attention gates to U-Net architectures to handle high-noise phase images (10), a technique highly relevant to our low-light screen scenario.

2 Project Plan and Novelty

2.1 Methodology

To ensure robustness against environmental noise and screen limitations, the proposed system will utilize a hybrid approach. The development pipeline is structured as follows:

1. **System Calibration:** We will determine the intrinsic parameters of the webcam and the extrinsic geometric relationship between the laptop screen and the camera. We will adopt the "screen-camera" calibration methods discussed by Vodilka et al. (2025), which allow for markerless scanning in confined spaces (4).
2. **Active Pattern Projection:** We will bypass static image projection in favor of a dynamic pattern engine. To improve robustness on shiny surfaces (specularity), we will implement the "Generative Asynchronous" projection strategy proposed by Lu et al. (2024), adapted for screen refresh rates (8).
3. **Reconstruction Optimization:** Captured frames will be processed to extract the wrapped phase. We will compare standard multi-frequency temporal unwrapping against a lightweight deep-learning-based unwrapping approach inspired by Evans et al. (2024) (9).

2.2 Novelty and Creative Elements

This project extends beyond standard implementations by specifically addressing "Perceptual Factors" and "Robustness" on low-end hardware:

- **Screen-Specific Optimization:** Unlike industrial scanners, laptop screens have gamma distortion and low brightness. We will implement a "Differential HDR" capture scheme to isolate the pattern from ambient light, similar to the High Dynamic Range techniques discussed by Zhang et al. (2023) (12).
- **Portable Form Factor:** By eliminating the external projector, we create a "Laptop-Only" scanner. This aligns with the trend of "Generalizable Fringe-to-Phase" frameworks that seek to democratize 3D scanning (11).
- **Robust Correspondenceless Matching:** We will validate our system on featureless white ceramics, a challenging benchmark where traditional stereo vision fails, utilizing the error analysis frameworks suggested by Wang et al. (2025) (5).

3 Milestones

- **Week 1-2 (Literature Setup):**
 - Review codebases for Single-Shot Structured Light (2).
 - Configure Python environment (OpenCV, PyTorch) and calibrate the webcam.
- **Week 3-4 (Pattern Engine):**
 - Develop the module for generating Phase-Shifted Sinusoids.
 - Implement the synchronized capture loop.
- **Week 5-6 (Reconstruction Algorithm):**
 - Implement Phase Wrapping and Unwrapping algorithms.
 - *Checkpoint:* First successful scan of a test object.
- **Week 7 (Refinement):**
 - Integrate "Differential HDR" logic for low-light performance.
 - Compare results against ground truth (measured object dimensions).
- **Week 8 (Final Report):**
 - Finalize experimentation and write the report.

4 Breakdown of Responsibilities

Team Member 1: Algorithm & Mathematics Lead

- Research and implement the Phase Wrapping and Phase Unwrapping algorithms (Spatial and Temporal).
- Develop the triangulation logic to convert unwrapped phase maps into dense 3D Point Clouds.
- Integrate "Deep Unrolling Networks" or similar deep-learning-based optimization techniques to improve reconstruction quality.
- Author the "Methodology" and "Algorithm Description" sections of the final report.

Team Member 2: Systems & Calibration Lead

- Perform geometric system calibration to determine the intrinsic (webcam) and extrinsic (screen-to-camera) parameters.
- Develop the Pattern Generation Engine (Python/OpenGL) to create precise Phase-Shifted Sinusoids and Gray Codes.

- Implement the synchronized capture pipeline to ensure perfect timing between screen projection and camera exposure.
- Author the "System Setup" and "Calibration" sections of the final report.

Team Member 3: Evaluation, Testing & Optimization Lead

- Design and conduct experiments on challenging objects (e.g., shiny surfaces, dark materials) to test system robustness.
- Implement the "Differential HDR" logic to improve scanning performance in low-light screen conditions.
- Perform quantitative error analysis by comparing scan results against ground-truth measurements.
- Author the "Experiments and Results" section and create the final presentation slides.

References

- [1] Y. Li, J. Xu, and Q. Zhang, "Fringe-Based Structured-Light 3D Reconstruction: Principles, Projection Technologies, and Deep Learning Integration," *Sensors*, vol. 25, no. 20, 6296, 2025.
- [2] R. Zuo et al., "Deep-learning-based 3D reconstruction from single-shot structured-light image," *Proceedings of SPIE*, vol. 13635, 2025.
- [3] H. Guo et al., "Deep-learning-enabled single-shot fringe projection profilometry based on inner shifting-phase encoding," *Optics Express*, vol. 33, no. 23, pp. 49530, 2025.
- [4] A. Vodilka et al., "Design of 3D Scanning Technology Using a Method with No External Reference Elements," *Applied Sciences*, vol. 15, no. 8, 4533, 2025.
- [5] H. Wang et al., "Line-Structured Light-Based Three-Dimensional Reconstruction Measurement System with an Improved Scanning-Direction Calibration Method," *Remote Sensing*, vol. 17, no. 13, 2236, 2025.
- [6] R. Zuo, S. Wei, and J. U. Kang, "Deep Learning-based Single-shot Fringe Projection Profilometry," in *Advanced Biomedical and Clinical Diagnostic and Surgical Guidance Systems XXII*, SPIE, 2024.
- [7] Z. Chen et al., "Unsupervised Deep Unrolling Networks for Phase Unwrapping," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [8] L. Lu et al., "Generative deep-learning-embedded asynchronous structured light for three-dimensional imaging," *Advanced Photonics*, vol. 6, no. 4, 046004, 2024.

- [9] R. G. Evans et al., “Deep Learning for Single-Shot Structured Light Profilometry: A Comprehensive Dataset and Performance Analysis,” *Journal of Imaging*, vol. 10, no. 8, 179, 2024.
- [10] M. Gontarz et al., “Phase unwrapping using deep learning in holographic tomography,” *Optics Express*, vol. 31, no. 12, pp. 18964-18992, 2023.
- [11] L. Yang et al., “Generalized Fringe-to-Phase Framework for Single-Shot 3D Reconstruction Integrating Structured Light with Deep Learning,” *Sensors*, vol. 23, no. 10, 4836, 2023.
- [12] S. Zhang et al., “High-dynamic-range 3D measurement based on deep learning and fringe projection,” *Optics and Lasers in Engineering*, vol. 160, 107297, 2023.